香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

2024-2025 Term 2

Midterm Examination

March 29th, 2025; Time Allowed: 3 Hours

NAME (print) _____

CUHKSZ ID _____

Room No. _____

Seat No. _____

- **Show all your work**. Correct answers with little supporting work will not be given credit.
- Closed Book Exam: One piece of double-sided A4 reference paper, a scientific calculator, and a paper-based dictionary are allowed.
- Unless approved by the instructors, students who arrive more than 30 minutes late will NOT be admitted.
- The total points are 120 points. You need to finish ALL the questions in 3 hours (180 minutes).

#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	TOTAL
10	10	10	10	10	10	10	10	10	10	10	10	120

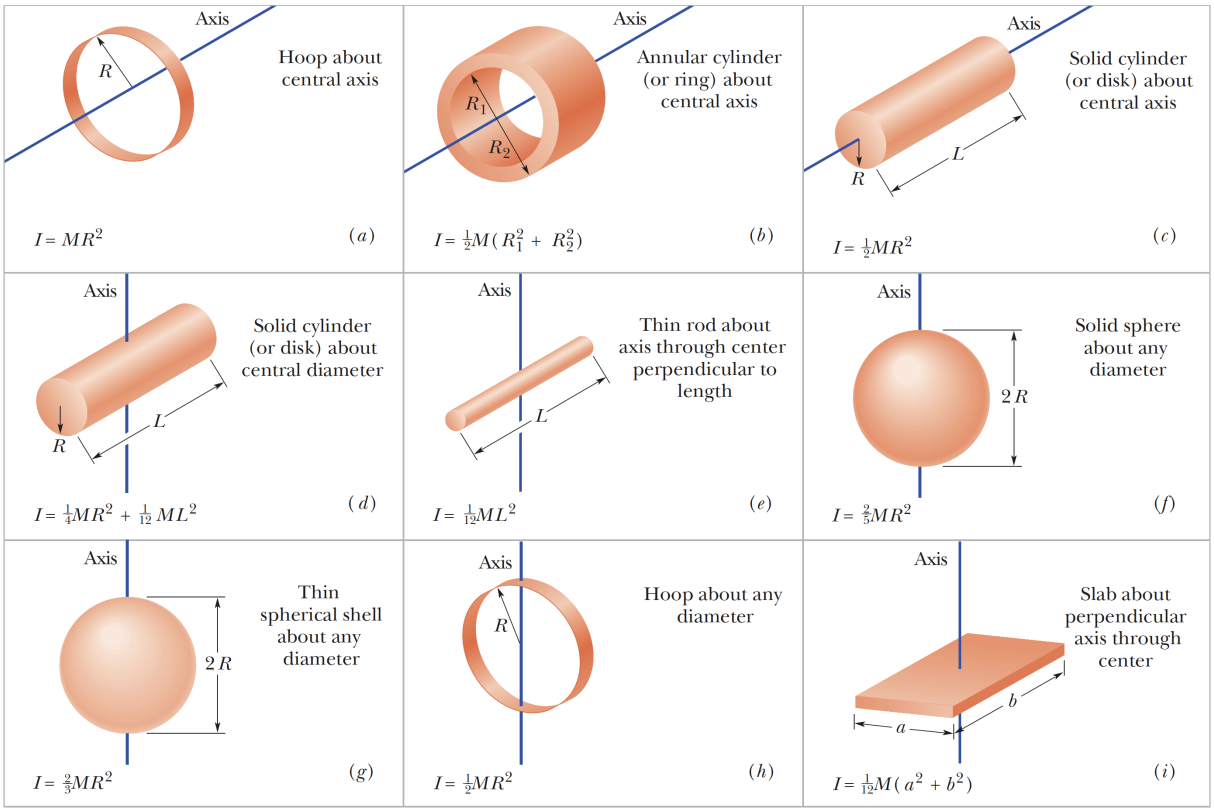
$g = 9.8\,\mathrm{m/s^2}$ for all questions that are involved with gravitational acceleration g.

Summary of Basic Calculus:

$$
\begin{aligned}
&e^{ix} = \cos x + i \sin x, \\
&\frac{d}{dx} x^n = n x^{n-1}, \\
&\frac{d}{dx} e^{ax} = a e^{ax}, \\
&\frac{d}{dx} \ln ax = \frac{1}{x}, \\
&\frac{d}{dx}(uv) = v\frac{d}{dx}u + u\frac{d}{dx}v, \\
&\int x^n dx = \frac{x^{n+1}}{n+1} + C, \ \ (n \neq -1), \\
&\int \frac{dx}{x} = \ln x + C, \\
&\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad C \text{ is a constant.}
\end{aligned}
$$

Taylor series:

$$
\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots (\text{all } x) \\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \cdots (\text{all } x) \\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + \cdots (\text{all } x).
\end{aligned}
$$



1. Consider two vectors $\vec{a} = 1.0\hat{i} + 2.0\hat{j} - 3.0\hat{k}$, and $\vec{b} = -3.0\hat{i} + 4.0\hat{j} + 2.0\hat{k}$. **(10 pts)**

(a) What is $(\vec{a} + \vec{b})$? **(2 pts)**

(b) What is $(\vec{a} \times \vec{b})$? **(2 pts)**

(c) What is $(\vec{a} \cdot \vec{b})$? **(2 pts)**

(d) Calculate $\cos\theta$, where θ is the angle between the two vectors $\vec{a}$ and $\vec{b}$. **(4 pts)**

2. If the position of a particle is given by

$$x(t) = 2.0t^2 + 3.0t + 1.0$$

$$y(t) = 4.0t + 2.0$$

where x and y are in meters, and t is in seconds. Solve the following:

(a) Position of the particle at $t = 1.0\,\text{s}$ in unit vector form. **(2 pts)**

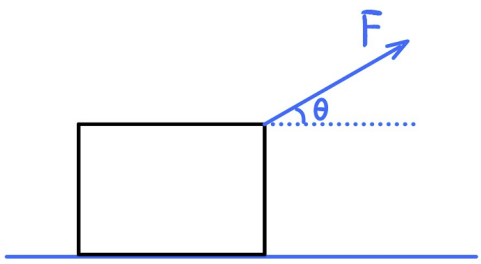
(b) Speed of the particle at $t = 1.0\,\text{s}$. **(2 pts)**

(c) Magnitude of acceleration of the particle at $t = 1.0\,\text{s}$. **(2 pts)**

(d) Magnitude of the displacement of the particle from $t = 0\,\text{s}$ to $t = 1.0\,\text{s}$. **(2 pts)**

(e) Average velocity of the particle from $t = 0\,\text{s}$ to $t = 1.0\,\text{s}$. **(2 pts)**
Hint: Indicate both the magnitude and direction of the average velocity. You may use $\tan^{-1}$ to represent the angle.

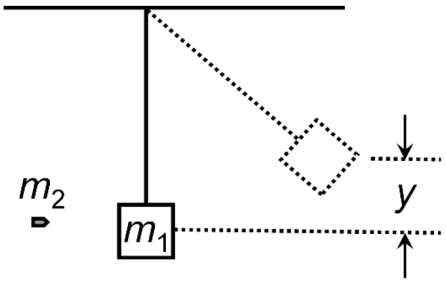
3. Suppose you are moving a crate on a horizontal floor with mass $m = 1\,\text{kg}$ by pulling upward on the rope at an angle of θ above the horizontal. **(10 pts)**



(a) Given the angle $\theta = 30^\circ$ and the coefficient of the kinetic friction $\mu = 0.20$, calculate the force applied to keep it moving with constant velocity? **(5 pts)**

(b) Can you find an angle where the required force is minimum? **(5 pts)**
Hint: You may use $\tan\theta$ to represent the angle.

4. A wooden block with mass m_1 is initially at rest, suspended by a massless string. A bullet with mass m_2 is fired horizontally toward the block. Upon collision, the bullet becomes embedded in the block, causing the combined mass to swing up and reach a maximum height y. Using the variables y, m_1, m_2, and the gravitational acceleration g. **(10 pts)**



(a) Derive an expression for the speed v_t of the combined mass right after the collision. **(4 pts)**

(b) Derive an expression for the initial speed v_0 of the bullet before the collision. **(3 pts)**

(c) Calculate the energy lost during the collision. **(3 pts)**

5. A particle moves along the x-axis under the influence of a force derived from a potential-energy function $U(x) = ax^2 - bx^3$, a and b are positive constants. **(10 pts)**

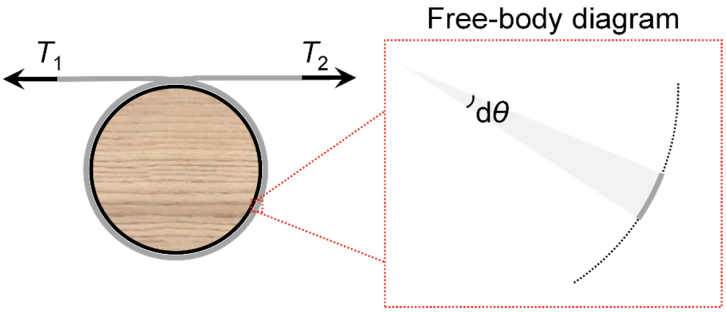
(a) Find the expression for the force $F(x)$ acting on the particle as a function of x. **(2 pts)**

(b) Determine the positions of equilibrium for the particle. **(4 pts)**

(c) Classify each equilibrium point as stable or unstable and provide reasons. **(4 pts)**

6. A massless rope is wrapped around a vertical pole, completing N full revolutions. The total angle of the wrap is $\theta = 2\pi N$. The coefficient of static friction between the rope and the pole is μ_s. The rope is pulled with a tension T_1 at one end, while the other end exerts a tension T_2 ($T_2 > T_1$). Assume the rope is in equilibrium, with just enough static friction to prevent it from sliding. **(10 pts)**

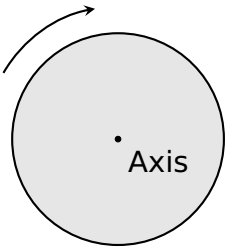
(a) Please draw the free-body diagram for a small piece of the rope that subtends an angle $d\theta$. **(3 pts)**



(b) Please derive an expression of T_2 in terms of T_1, μ_s, and N. **(5 pts)**

(c) If T_2 keeps increasing, please describe how the rope would behave, including the changes in frictional force and the overall motion of the rope. **(2 pts)**

7. A solid disk of mass 5.0 kg and radius 0.25 m starts from rest and is brought to an angular speed of 30 rad/s by a motor in 10 s. The angular acceleration is constant. Please answer the following questions with results of two significant figures. **(10 pts)**



(a) What is the magnitude of the angular acceleration of the disk? **(2 pts)**

(b) How many radians does the disk rotate during this time? **(2 pts)**

(c) What is the increase of rotational kinetic energy of the disk during the 10-second period? **(4 pts)**

(d) What is the average power input from the motor to the disk during the 10-second period? **(2 pts)**

8. A pulsar is a rotating neutron star that emits pulses. Based on the pulses, we can measure its rotation period $T = T_0 + \beta(t) \cdot (t - t_0)$. Suppose the angular acceleration α is a constant, which means $\beta(t)$ is not a constant. When $t = t_0$, the measurements tell us that $T = T_0$ and $\beta(t) = \beta_0$. **(10 pts)**

   (a) What is the magnitude of pulsar's angular velocity ω_0 at t_0? Keep π in your results. **(2 pts)**

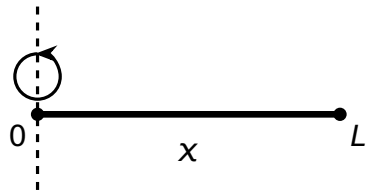
   (b) The pulsar can be assumed to be a uniform solid sphere with radius R and mass m. What is the pulsar's rotational kinetic energy at t_0? **(3 pts)**

   (c) Find the magnitude of the constant angular acceleration α using T_0 and β_0. **(5 pts)**

   Hint: Use chain rule $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ to derive α.

9. A thin rod of length L lies along the x-axis, with one end at the origin ($x = 0$) and the other end at $x = L$. The rod has a non-uniform linear mass density given by:

$$\lambda(x) = \lambda_0 \left(1 + \frac{x}{L}\right).$$



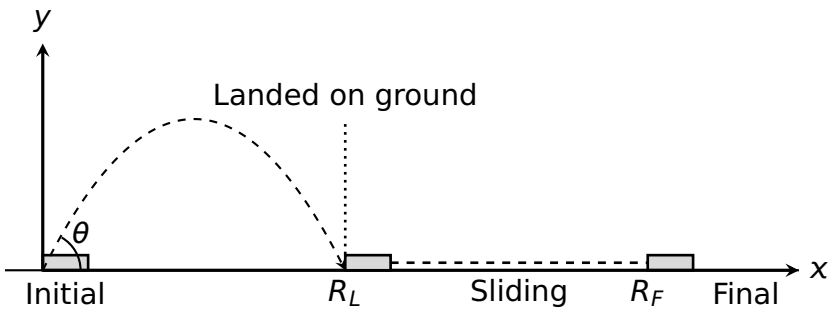
(10 pts)

(a) Calculate the total mass M of the rod. **(5 pts)**

(b) The rod rotates about an axis perpendicular to the rod and passing through the origin ($x = 0$). The moment of inertia I can be written in the form of cML^2. Find c. **(5 pts)**

10. A brick is thrown from ground level with initial velocity v_0, at an angle θ with respect to the (horizontal) ground. Assume that: **(10 pts)**

- The long face of the brick remains parallel to the ground at all times.
- There is no deformation in the ground or the brick when the brick hits the ground.
- The coefficient of friction between the brick and the ground is μ.
- The brick does not bounce upon hitting the ground. That is to say that the brick collides with the ground inelastically in the vertical direction.



(a) Find the range R_L of the brick when it lands on the ground. **(1 pts)**

(b) Find the velocity of the brick in the x and y directions right before landing. **(2 pts)**

(c) Find the velocity of the brick in the x and y directions right after landing, assuming $\mu \leq 1$.
Hint: The brick slows down when it hits the ground. Think in terms of impulse. **(2 pts)**

(d) Find total horizontal distance R_F. Suppose the horizontal movement during the collisiion with ground is negligible. **(2 pts)**

(e) Determine the angle θ such that the brick travels the maximum total horizontal distance R_F before finally coming to rest. **(2 pts)**

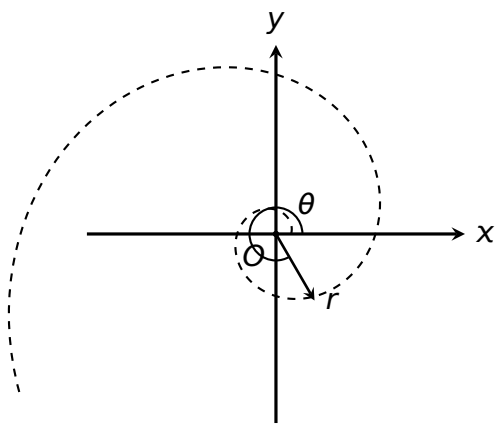
(f) If $\mu > 1$, what would be the angle θ such that the brick travels the maximum total horizontal distance R_F? **(1 pts)**

Hint: What is the brick's horizontal velocity after it hits the ground when $\mu > 1$?

11. A bird flies on a spiral trajectory such that its polar coordinates at time t are given by

$$r = Re^{\omega t}, \quad \theta = \omega t,$$

where R and ω are positive constants. **(10 pts)**



(a) Find $x(t)$ and $y(t)$. **Hint:** $x = r\cos\theta$, $y = r\sin\theta$. **(2 pts)**

(b) Find $v_x(t)$ and $v_y(t)$. **(2 pts)**

(c) Find $a_x(t)$ and $a_y(t)$. **(2 pts)**

(d) Show that the angle between $\vec{r} = x\hat{i} + y\hat{j}$ and $\vec{v}$ is always $\pi/4$. **(2 pts)**

(e) Find the angle between $\vec{r}$ and $\vec{a}$. **(2 pts)**

12. Moments of Inertia and Scaling Laws. **(10 pts)**

The purpose of this problem is to use scaling laws first discovered by Galileo, along with the parallel axis theorem, to calculate the moments of inertia of various objects.

For a uniform object in n dimensions, one finds that $m = \rho \int \mathrm{d}^n r$. Thus, another object with half the size would have the mass

$$m_{1/2} = \frac{1}{2^n} m.$$

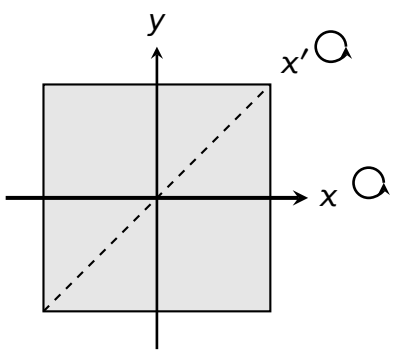
Similarly, the corresponding moment of inertia $I = \rho \int x^2 \mathrm{d}^n r$ would scale like

$$I_{1/2} = \frac{1}{2^{n+2}} I.$$

DO NOT PANIC! Let us give you some examples to understand the statement above.

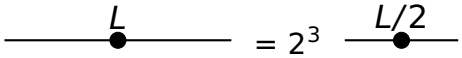
- For one dimension ($n = 1$)
  - $m = \rho \int \mathrm{d}r$, and $m_{1/2} = \frac{1}{2} m.$; $I = \rho \int x^2 \mathrm{d}r$, and $I_{1/2} = \frac{1}{2^3} I.$
- For two dimensions ($n = 2$)
  - $m = \rho \int \mathrm{d}^2 r$, and $m_{1/2} = \frac{1}{2^2} m.$; $I = \rho \int x^2 \mathrm{d}^2 r$, and $I_{1/2} = \frac{1}{2^4} I.$
- For three dimensions ($n = 3$)
  - $m = \rho \int \mathrm{d}^3 r$, and $m_{1/2} = \frac{1}{2^3} m.$; $I = \rho \int x^2 \mathrm{d}^3 r$, and $I_{1/2} = \frac{1}{2^5} I.$

(a) Consider a two-dimensional ($n = 2$) uniform square with a length of each side L, as shown below. Find the moment of inertia about the x-axis (I_x). (Use any method you prefer.) **(1 pt)**

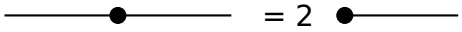


(b) Find the moment of inertia about the x' (diagonal) axis ($I_{x'}$). (Use any method that you want.) **(1 pt)**

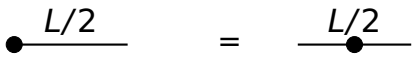
(c) Let us illustrate the scaling technique in the pictures below with a 1-dimensional rod of total length L and mass M. If one denotes the moment of inertia of an object by a picture of the object, with a dot signifying the axis, then one has the following form.



Scaling law with $n = 1$



Additivity

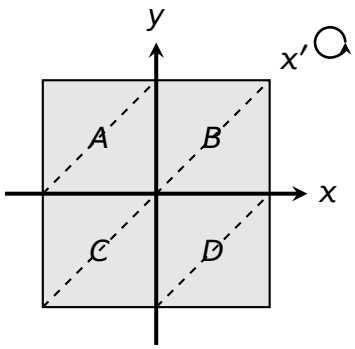


$$+ \frac{M}{2}\left(\frac{L}{4}\right)^2$$

Parallel axis theorem

Using the above three equations, <u>find</u> the moment of inertia for a thin rod with mass M and length L, which corresponds to the symbol on the left-hand side of the first equation. **(1 pt)**

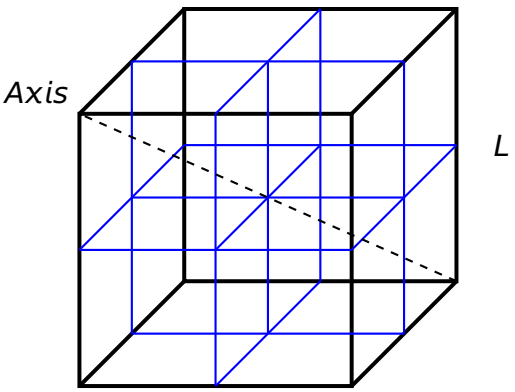
(d) Now consider the rotation about the x' (diagonal) axis again for a 2-dimensional ($n = 2$) uniform square with a length of each side L, as shown below. **(3 pts)**



Use the scaling technique outlined above for the one-dimensional case to redo part (b) and compute the moment of inertia about the x' axis by following the steps below:

- Divide the square into four smaller squares with half the size: A, B, C, and D.
- Write down the corresponding three equations that give the correct answer for $I_{x'}$. Each equation is worth 1 point.

(e) Using the technique outlined above, find the moment of inertia of a uniform cube with length L and mass M rotating about the diagonal axis (dashed line) illustrated below.



Again, you need to write down three equations, each given 1 point. The final correct answer is also worth 1 point. **(4 pts)**